

Information Retrieval Assignment Report-

HW2

This assignment is for implementing a custom inverted index to support information retrieval on the Cranfield dataset. The system replaces Elasticsearch and provides full control over indexing and ranking processes.

Two indexes were constructed:

- **Unstemmed index:** stopwords removed, no stemming
- **Stemmed index:** stopwords removed, Porter2 stemming applied

The system supports tokenization, stopword removal, stemming, and multiple ranking models.

1 Preprocessing

- **Tokenization:** Only letters and numbers are allowed; periods appear only within numbers (e.g., 98.6)
- **Lowercasing:** All text converted to lowercase
- **Stopword Removal:** 127 NLTK stopwords removed
- **Stemming:** Porter2 algorithm applied (for the stemmed index only)

2 Inverted Index

Each term in the index stores:

- Document Frequency (DF)
- Total Term Frequency (TTF)
- Postings list (docID, TF, positions)

3 Ranking Models

Three ranking models were implemented:

- **Okapi TF** – considers term frequency and document length
- **BM25** – extends Okapi TF with IDF weighting
- **Proximity Model** – ranks documents based on closeness of query terms

4 Results

4.1 Index Statistics

Index	Documents	Unique Terms
Unstemmed	1,400	7,667
Stemmed	1,400	4,947

Stemming reduced vocabulary size by **35.5%**.

4.2 MAP Scores

Model	MAP Score
BM25 (Unstemmed)	0.008124
BM25 (Stemmed)	0.008041
Okapi TF (Unstemmed)	0.007254
Okapi TF (Stemmed)	0.007100
Proximity	0.006599

Conclusion

BM25 (Unstemmed) achieved the best performance, demonstrating the effectiveness of IDF weighting in ranking relevant documents. Stemming slightly reduced performance (by about 1–2%) because the Cranfield dataset contains specialized terms where different word forms carry distinct meanings. Compared to Okapi TF, BM25 consistently performed better due to its ability to emphasize informative terms. The Proximity model showed the lowest performance, mainly due to short queries and the dispersed occurrence of terms within documents.